Hydration of Cement

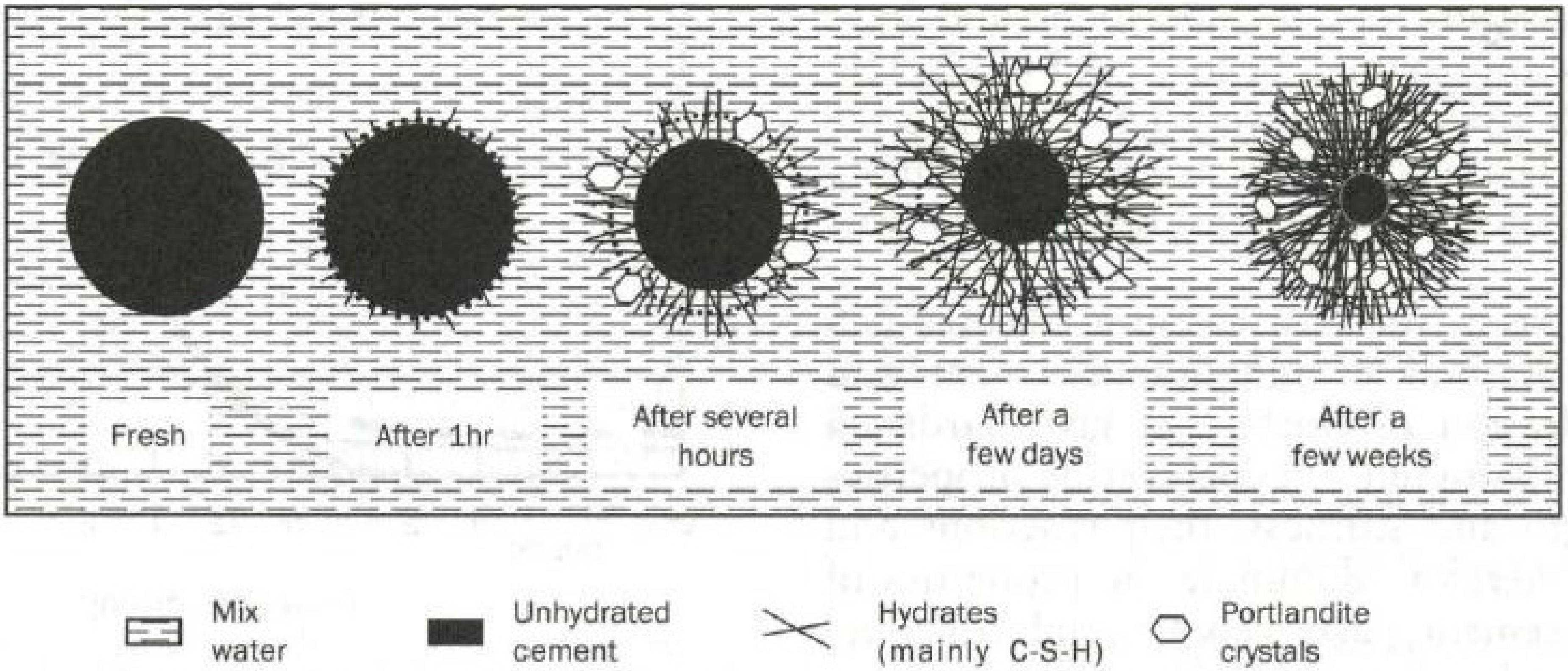
Reaction between cement particles and water

- Change in matter
  - Cement Hydration Products formed
  - Decrease in porosity
- Change in energy level
  - Heat is generated; Temperature rise
- Rate of reaction
  - Composition and Fineness of cement
  - Increased with the increase in temperature
  - Decrease with the increase in time
  - Decrease with the decrease in moisture content

When Water Is Added to Cement, What Happens?

- Dissolution of cement grains
- Growing ionic concentration in "water" (now a solution)
- Formation of compounds in solution
- After reaching a saturation concentration, compounds precipitate out as solids ("hydration products")
- In later stages, products form on or very near the surface of the anhydrous cement

Hydration of A Cement Grain With Time



Hydration of Calcium Silicates (75% of Cement)

- $2C_3S + 6H \rightarrow C_3S_2H_3 + 3CH \quad \Delta H = 500$ J/g

  $2C_2S + 4H \rightarrow C_3S_2H_3 + CH \quad \Delta H = 250$ J/g

  C-S-H

  - C_3S hydration is more rapid and evolves more heat
  - C_3S is responsible for the setting of cement (or concrete) and contributes to early age strength (2-3 hrs to 14 days)

    C_2S hydration occurs more slowly and contributes to later age strength after ~7-14 days

    C_3S produces more CH (source for sulfate attack, a durability concern)

Hydration of Tricalcium Aluminate

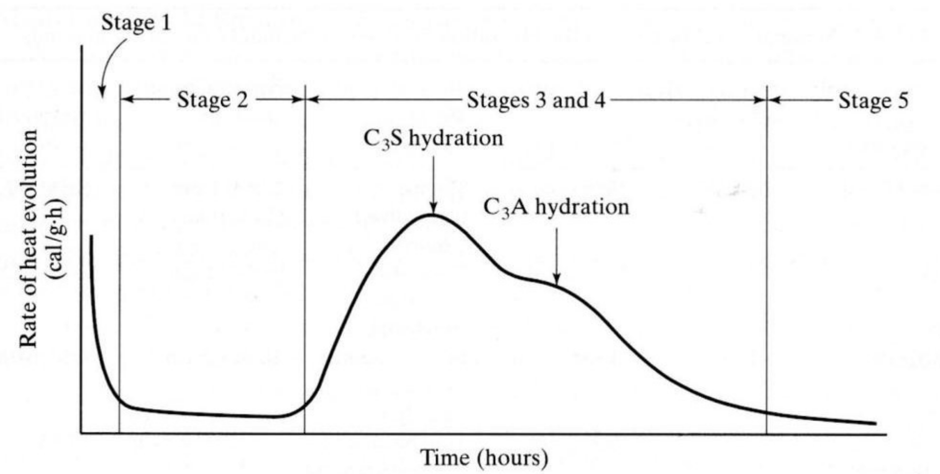
- Pure C_3A hydrates rapidly and produce flash set
  - $C_3A + 6H \rightarrow 3C_3AH_6$

- In the presence of gypsum, C_3A hydration is controlled and flash setting is avoided
  - $\underset{\text{}}{C_3A} + \underset{\text{Gypsum}}{3C\underline{S}H_2} + 26H \rightarrow \underset{\substack{\text{Calcium sulfoaluminate} \\ \text{(Ettringite)}}}{C_6A\underline{S}_3H_{32}}$ $\qquad \Delta H = 850$ J/g

- Additional C_3A is a source for sulfate attack
- The aluminates hydrates faster than silicates with increased heat of hydration

Heat of Hydration of Cement

- The hydration reaction of Portland cement are all exothermic, meaning liberates heat
- Quantity of heat evolved depends on
  - Chemical composition
  - Temperature
  - Age of cement



- Heat evolved (J/g)

$$H = a\,.\,C_3S + b.\,C_2S + c.\,C_3A + d.\,C_4AF$$

a (= 502J/g), b (= 260J/g), c (= 867J/g), d (= 419J/g)

are heat of hydration for the respective compounds

Oxide	Cement A	Cement B	Cement C
Silica	22.4	25.0	20.7
Lime	68.2	61.0	64.2
Iron	0.3	3.0	5.3
Alumina	4.6	4.0	3.9
SO3	2.4	2.5	2.0
Free lime	3.3	1.0	1.5

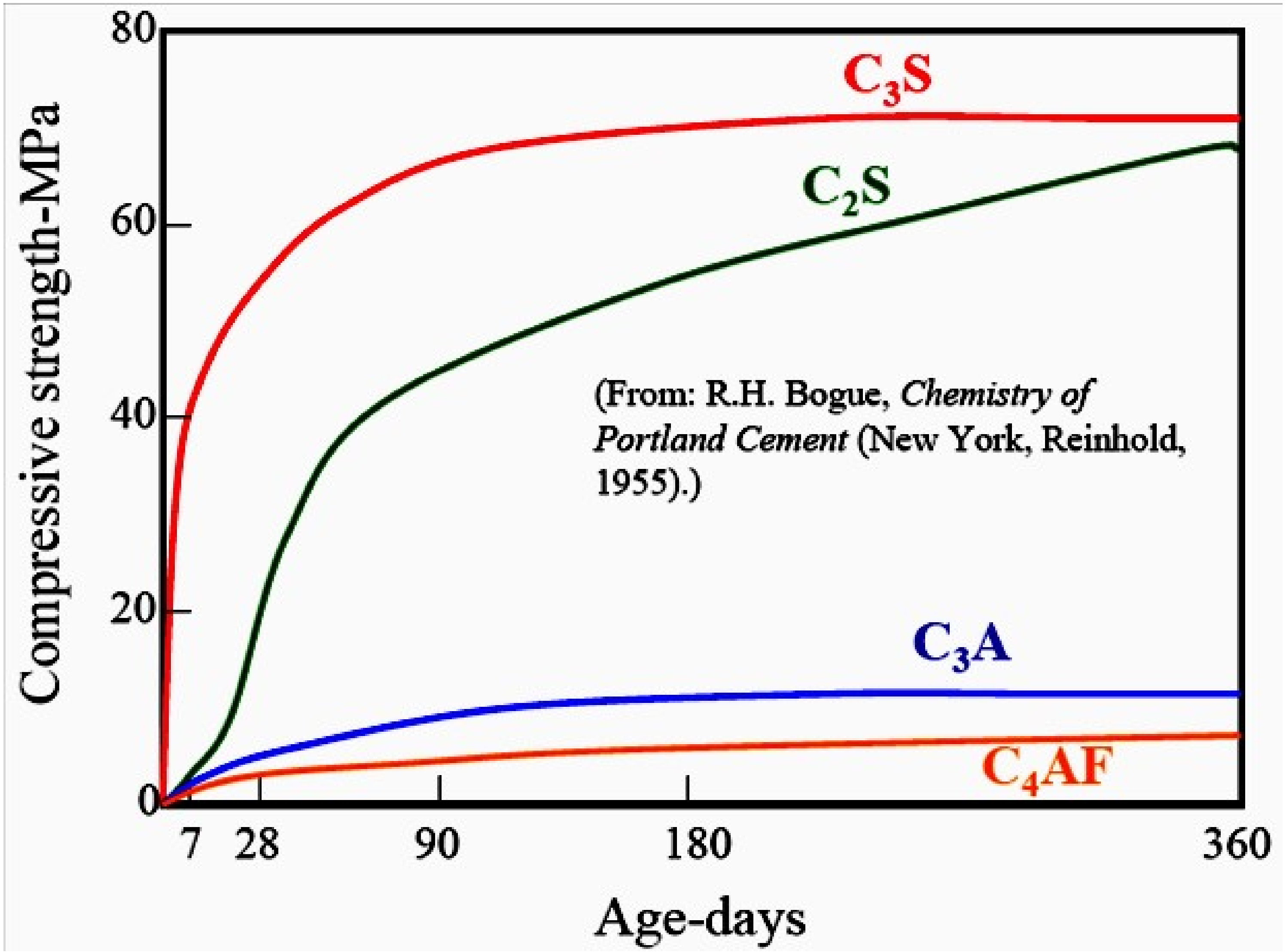
Using Bogue's equation, determine the major chemical compounds of these cements and highlight the expected performance of the concrete mixes of a given composition when these cements were used, with reference to the following: (i) rate of strength development; (ii) sulfate attack; and (iii) temperature increase in a thick concrete section.

(b) Assuming the heat of hydration of C_3S, C_2S, C_3A, and C_4AF are 500, 260, 860, and 420 kJ/kg, respectively, determine the heat of hydration for these cements

Major compounds contents (%) of three cements

Compound	Cement A	Cement B	Cement C
C_3S	69.15	19.98	64.49
C_2S	12.42	56.77	11.04
C_3A	11.68	5.53	1.38
C_4FA	0.91	9.12	16.11

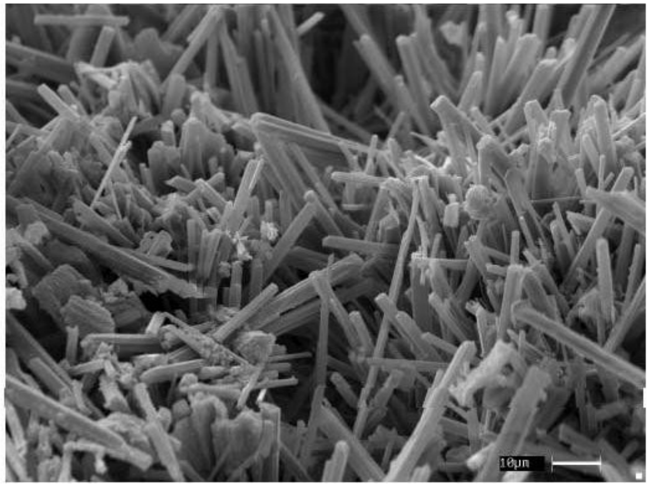
Strength development of major chemical compounds



Microstructure of Hydration Products

C-S-H

Calcium Hydroxide



Ettringite

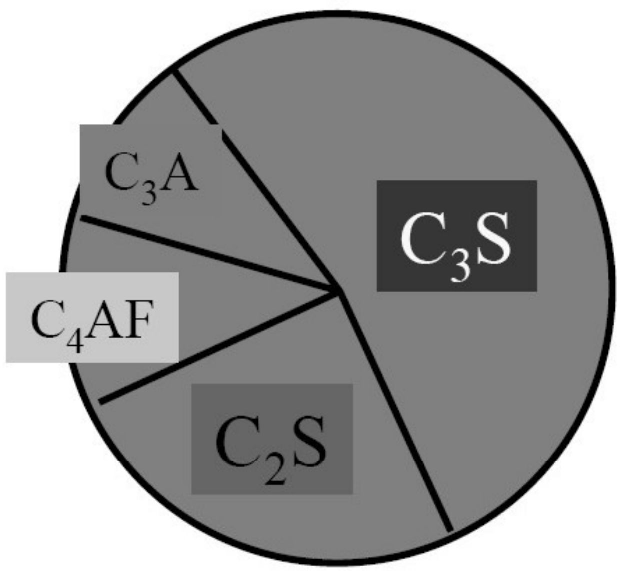
Photo courtesy of Paulo Monteiro, UC-Berkely

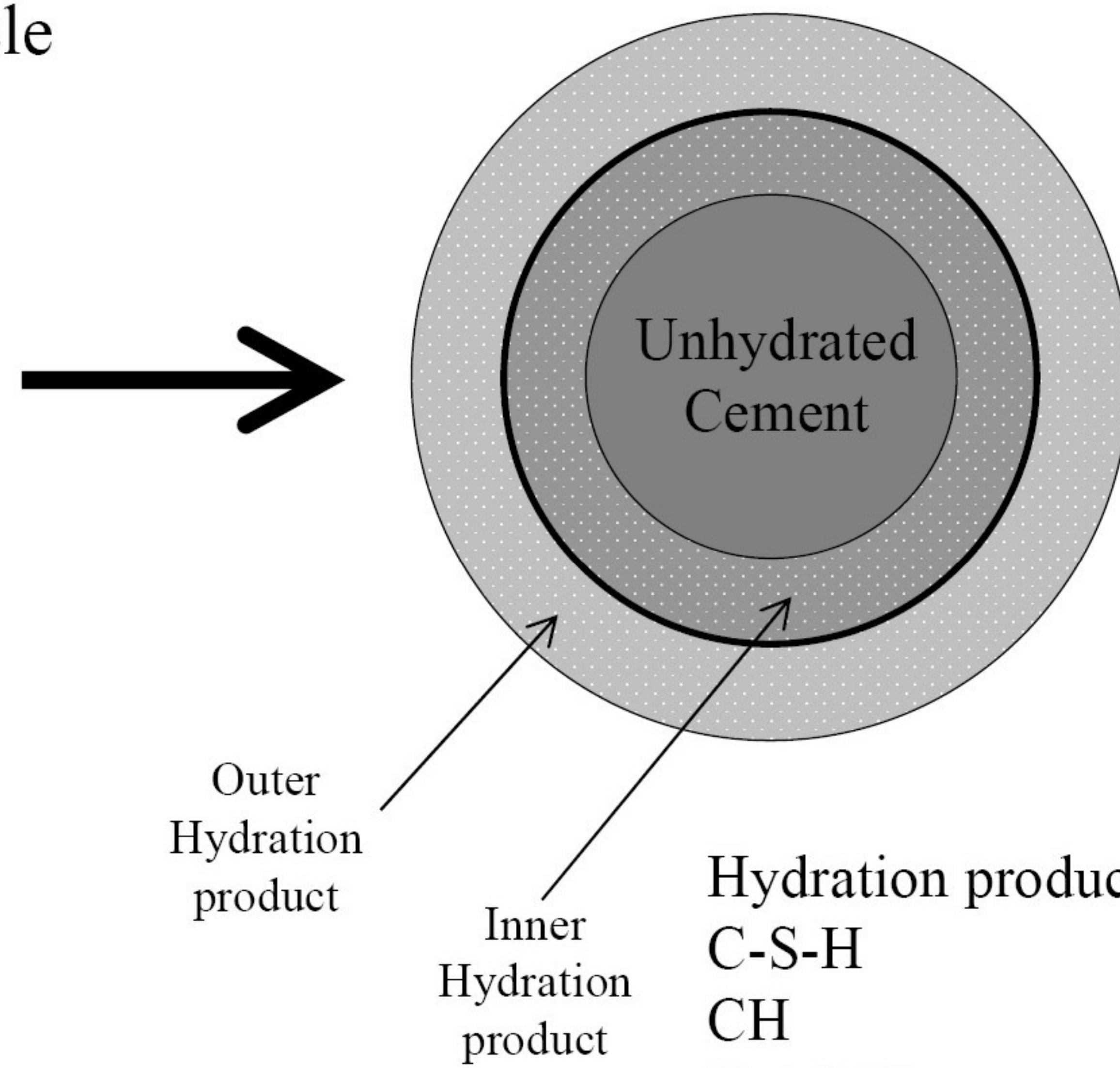
Characteristics of Hydration Products

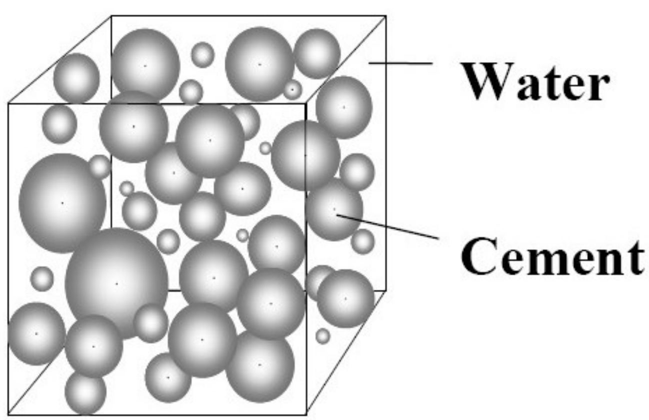
Phase	% by Vol.	Strength	Remarks
Calcium Silicate Hydrate (C-S-H)	50 – 60	Provides major cohesive force	
Calcium Hydroxide (CH)	20 – 25	Reduces porosity	Causes sulfate attack
Calcium Sulfoaluminates ($C_6A\underline{S}_3H_{32}$)	15 – 20	Not significant	Causes sulfate attack
Unhydrated cement	~ 5	Significant only in low porosity pastes	

Volume Changes During Hydration

Original cement particle



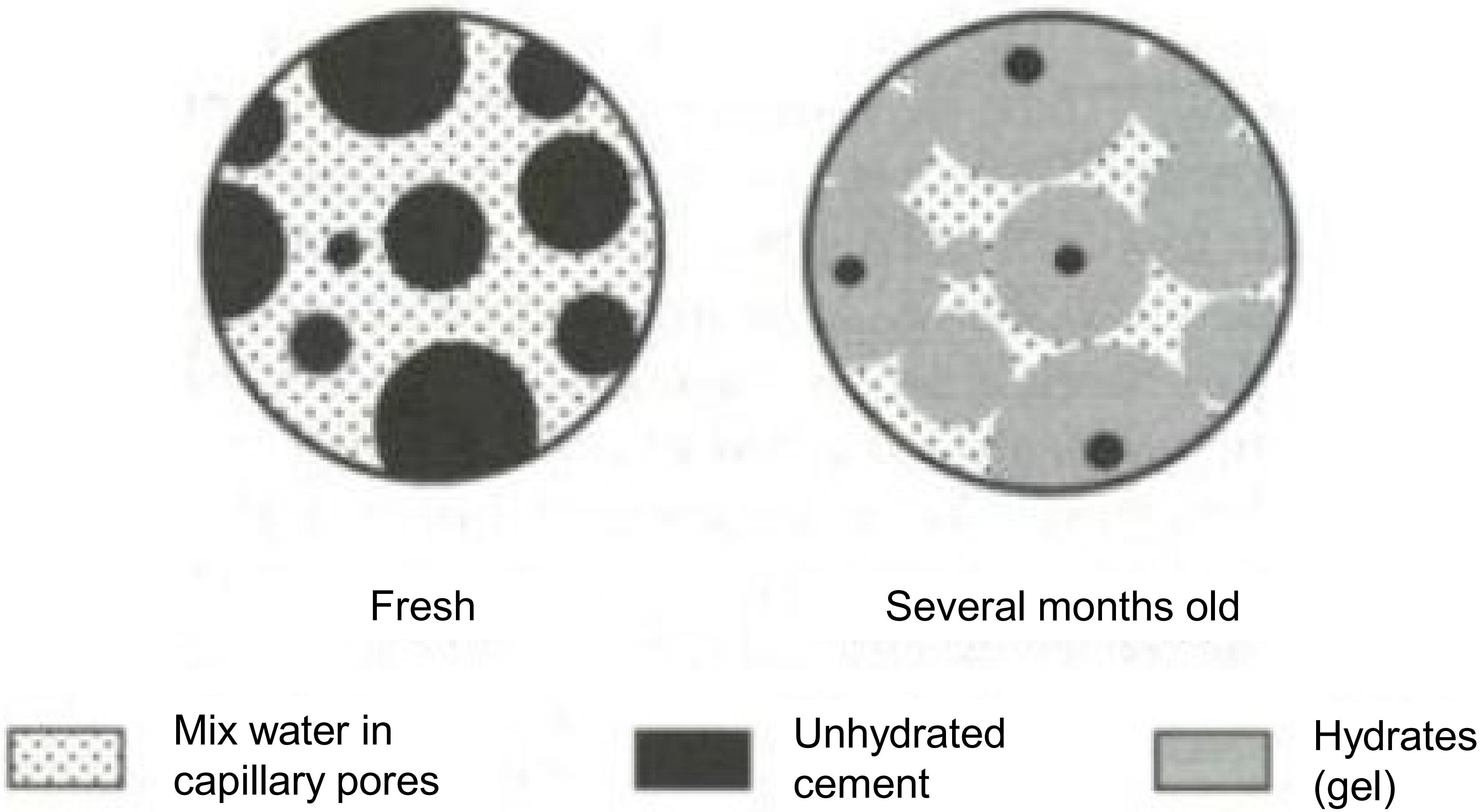




Hydration products:
C-S-H
CH
$C_6A\underline{S}_3H_{32}$
$C_4A\underline{S}H_{12}$

Volume of hydration product = 2.14x volume of cement

Structure of Cement Paste



Structure of Hydrated Cement Paste